Sub Spaces... class Notes

Friday, June 5, 2026 10:08 AM

Example 1: If V is any vector space and if $W = \{\vec{0}\}$ is the subset of V that consists of the zero vector only, then W is closed under addition and scalar multiplication since

is Subspace

$\vec{u} = \{0\}$

$\vec{v} = \{0\}$

(i) $u+v = \{0\} + \{0\} = \{0\} \in W$

(ii) $Ku = K\{0\} = \{0\} \in W$

Example 2:

$$W = \{(x, y); x \geq 0, y \geq 0\}$$

Prove and disprove W is subspace of R^2.

$u = (x_1, y_1)$, $x_1, y_1 \geq 0$ $\quad$ $v = (x_2, y_2)$, $x_2, y_2 \geq 0$

$u+v = (x_1, y_1) + (x_2, y_2)$

$(x_1+x_2, \; y_1+y_2)$; $x_1+x_2 \geq 0$, $y_1+y_2 \geq 0$

Closed under Addition

$Ku = K(x_1, y_1) = (Kx_1, Ky_1)$ if $K>0$ Then it is True

Let $K=-1$, $(-x_1, -y_1) \notin W$,

Let $u=(3,4)$, $v=(1,2)$ $\quad$ $u+v = (4,6) \in W$

$Ku = K(3,4)$, if $K=-1$, $(-3,-4) \notin W$

"W is NOT a Subspace"

Example 3:

Example 3:

Which of the following subset of R^2 with usual operations of addition and scalar multiplication are subspaces?

a) $W_1 = \{(x, y); x \geq 0\}$

b) $W_2 = \{(0, y); y \in R\}$

a) $u = (2, -3)$, $v = (5, 3)$

$u + v = (7, 0)$

$Ku = \boxed{(-2, 3)} \notin W$ $K = -1$

W_2

Let $u = (0, 5)$, $v = (0, -7)$

(i) $u + v = (0+0, \quad 5-7) = (0, -2) \in W_2$

(ii) $Ku = K(0, 5) = (0, 5K)$ $K = -1$

$(0, -5) \in W_2$

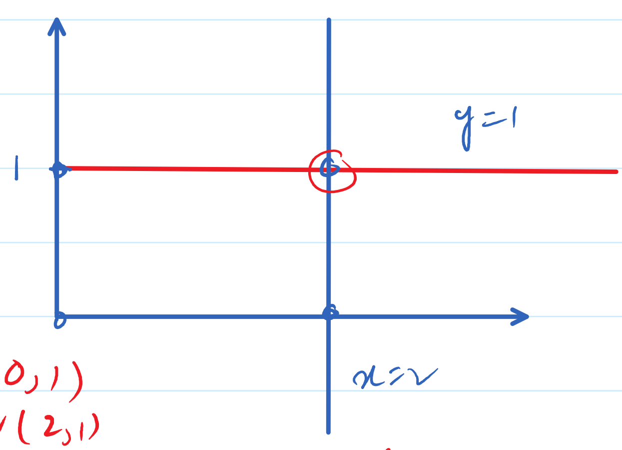
$W_2 \Rightarrow$ vector space.

Draw a Rectangle $0 \leq x \leq 2$, $0 \leq y \leq 1$

Prove/disprove this Rectangle is a subspace

$u = (0, 1)$, $v = (2, 0)$

$u + v = (2, 1)$



$u(0, 1)$

$v(2, 1)$

$u + v = (2, 2) \notin$ Rect

$Ku = K(0, 1)$

$K = -1$ $-1u = -1(0, 1) = (0, -1) \notin$ Rectangle

Example 5.

Which of the given subsets of the vector space M_{23} of all matrices are subspace? The set of all matrices of the form?

$$W_1 = \left\{ \begin{bmatrix} a & b & c \\ d & 0 & 0 \end{bmatrix}; where\ b = a + c \right\}$$

$U = \begin{pmatrix} 1 & 4 & 3 \\ 2 & 0 & 0 \end{pmatrix}$ $V = \begin{pmatrix} 5 & 8 & 3 \\ 2 & 0 & 0 \end{pmatrix}$

$U + V = \begin{pmatrix} 6 & 12 & 6 \\ 4 & 0 & 0 \end{pmatrix} \in W_1$

$-1\,U = \begin{pmatrix} -1 & -4 & -3 \\ -2 & 0 & 0 \end{pmatrix} \in W_1$

$W_1 =$ Subspace

a) $W_2 = \left\{ \begin{bmatrix} a & b & c \\ d & 0 & 0 \end{bmatrix}; where\ c > 0 \right\}$

a) $W_3 = \left\{ \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}; where\ a = -2c, f = 2c + d \right\}$

Q1. Which of the following subset of R^3 with usual operations of addition and scalar multiplication are subspaces?

Over to you

a) W_1 = All vectors of the form $(a, 0, 0)$

b) W_2 = All vectors of the form $(a, 1, 1)$

c) W_3 = All vectors of the form (a, b, c) where $b = a + c$.

d) W_4 = All vectors of the form (a, b, c) where $b = \underline{a + c} + 1$.

e) W_5 = All vectors of the form $(a, b, 0)$.

$u = (2, 7, 4)$

$v = (3, 9, 5)$

$u + v = (5, 16, 9)$

$\underline{9 + 5 + 1} \neq 16$

2. Use Theorem 4.2.1 to determine which of the following are subspaces of M_{nn}.

(a) The set of all diagonal $n \times n$ matrices.

(b) The set of all $n \times n$ matrices A such that $\det(A) = 0$.

(c) The set of all $n \times n$ matrices A such that $\mathrm{tr}(A) = 0$.

(d) The set of all symmetric $n \times n$ matrices.

(e) The set of all $n \times n$ matrices A such that $A^T = -A$.

(f) The set of all $n \times n$ matrices A for which $A\mathbf{x} = 0$ has only the trivial solution.

(g) The set of all $n \times n$ matrices A such that $AB = BA$ for some fixed $n \times n$ matrix B.

$$\begin{array}{ccc} + & - & + \\ \end{array}$$

$$\begin{pmatrix} 2 & 4 & 6 \\ 6 & 9 & 12 \\ 6 & 7 & 10 \end{pmatrix} = 2\begin{vmatrix} 9 & 12 \\ 7 & 10 \end{vmatrix} - 4\begin{vmatrix} 6 & 12 \\ 6 & 10 \end{vmatrix} + 6\begin{vmatrix} 6 & 9 \\ 6 & 7 \end{vmatrix}$$

$= 2(6) - 4(-12) + 6(42 - 63)$

$= 12 + 48 + 6(-21)$

$= 60 - 126 \neq$

$1(36 - 30) \times 2$

$4(5) \; -1(6)$
$+2(-7)$

$1(36-30)$

$4(5)$
$+2(-7)$
$20-6-14$

$$u = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{pmatrix} ; \begin{pmatrix} 4 & 1 & 2 \\ 5 & 3 & 1 \\ 9 & 4 & 3 \end{pmatrix}$$

$$= \begin{pmatrix} 5 & 3 & 5 \\ 7 & 7 & 7 \\ 12 & 10 & 12 \end{pmatrix} = 5(14) - 3(0) + 5(70-84)$$

$$= 70 \quad -70$$

$$= 0$$

$$A = \begin{pmatrix} -2 & 3 \\ 4 & 2 \end{pmatrix} \quad B \begin{pmatrix} -4 & 2 \\ -1 & +4 \end{pmatrix} \quad -1A : \begin{pmatrix} 2 & -3 \\ -4 & -2 \end{pmatrix}$$

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$$A+B = \begin{pmatrix} -6 & 5 \\ +3 & 6 \end{pmatrix}$$

Day = Thursday

Date : 11-06-2026

Venue = A-101

Menu = Subspace

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